

Practical 2: Estimation of Project Metrics for Pharmacy Management Software Project

Aim: To estimate project metrics for the Pharmacy Management System using the COCOMO (Constructive Cost Model) and to understand the advantages and drawbacks of the model.

Introduction: Project cost estimation is essential in software development to predict the resources, time, and budget required. The COCOMO model is a well-known cost estimation technique that uses project size and complexity to estimate effort and duration.

Objective:

- To estimate project parameters using the COCOMO model.
- To understand the advantages and limitations of COCOMO.
- To perform project cost estimation.

Project Estimation Technique: COCOMO Model The COCOMO (Constructive Cost Model) is a cost estimation model developed by Barry Boehm. It helps estimate the number of person-months required to develop a software project based on the size measured in KLOC (Thousands of Lines of Code).

Basic COCOMO Model: The basic model estimates effort (in person-months) using the formula:

$$\text{Effort (Person-Months)} = a * (\text{KLOC})^b$$

Where,

- **KLOC:** Thousands of Lines of Code
- **a, b:** Constants based on the project type (organic, semi-detached, or embedded).

Project Categories:

- **Organic:** Simple projects with small teams and flexible requirements.
- **Semi-Detached:** Projects with moderate complexity.
- **Embedded:** Complex projects with stringent hardware, software, and operational constraints.

Example: For a Pharmacy Management System estimated at 50 KLOC with constants ($a = 2.4$, $b = 1.05$):

$$\text{Effort} = 2.4 * (50)^{1.05}$$

$$\text{Effort} \approx 130 \text{ Person-Months}$$

Advantages of COCOMO:

- **Simplicity:** Easy to use and understand.
- **Scalability:** Suitable for small to large projects.
- **Structured:** Provides a systematic approach to cost estimation.
- **Customization:** Parameters can be adjusted based on project-specific factors.

Drawbacks of COCOMO:

- **Accuracy Issues:** Basic COCOMO is less accurate for complex projects.
- **Static Model:** Assumes a fixed project structure, which may not reflect real-world changes.
- **Data Dependency:** Requires accurate KLOC estimation, which is difficult at the early stages.

Project Cost Estimation: Cost estimation involves calculating the total resources and time required to complete the project. In COCOMO, cost is derived by calculating the number of person-months and multiplying it by the cost per person-month.

Formula: Total Cost = Effort (Person-Months) * Cost per Person-Month

Conclusion: Using the COCOMO model for cost estimation provides a structured approach to predicting project metrics for the Pharmacy Management System. Despite its limitations, it helps in resource planning, budgeting, and setting realistic project timelines.